

Measurement of the Imperceptible Threshold for Color Vibration Pairs Selected by using MacAdam Ellipse

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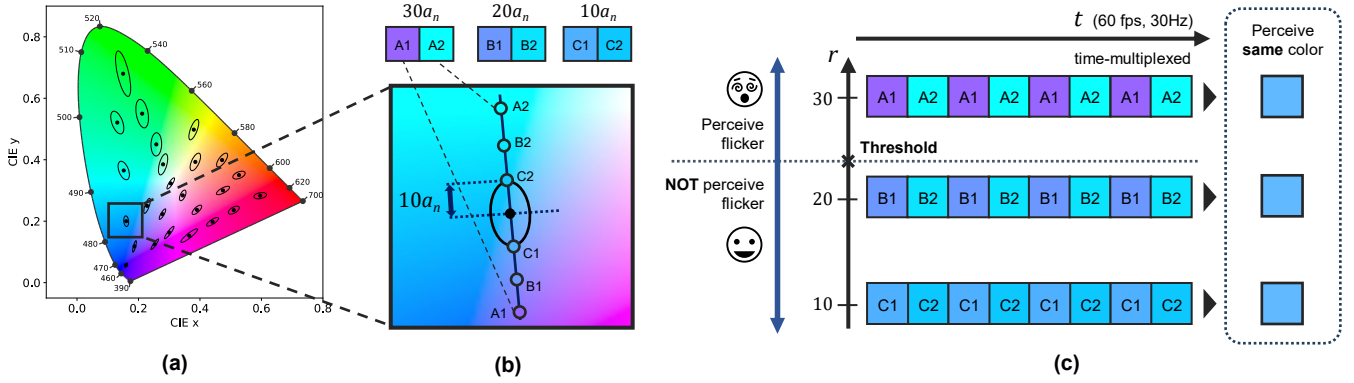


Figure 1: (a) MacAdam Ellipse [MacAdam 1942], the region that encompasses all colors that cannot be distinguished by the average human eye from the color in the center of the ellipse on the xy chromaticity diagram. For illustration, the ellipses are shown 10 times their actual size. (b) The two points of each MacAdam ellipse, whose major diameter a_n is multiplied by r , are selected for color vibration. (c) Users view the color vibration on a color-calibrated display, where the two colors selected by (b) are displayed alternately. In every pair, the human eye perceives the intermediate color resulting from the fusion of the pair.

ABSTRACT

We propose an efficient method for searching for color vibration pairs that are imperceptible to the human eye based on the MacAdam ellipse, an experimentally determined color-difference range that is indistinguishable to the human eye. We created color pairs by selecting eight colors within the sRGB color space specified by the ellipse, and conducted experiments to confirm the threshold of the amplitude of color vibration amplitude at which flicker becomes imperceptible to the human eye. The experimental results indicate a general guideline for acceptable amplitudes for pair selection.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Mixed / augmented reality**.

KEYWORDS

color perception, imperceptible color vibration, MacAdam ellipse

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1 INTRODUCTION

Imperceptible color vibration is a perceptual phenomenon where the human visual system interprets the rapid alternating presentation of two colors with the same luminance but different chromaticity as a single fused intermediate color [Jiang et al. 2007]. This effect occurs when the alternation frequency exceeds the critical color fusion frequency (CCFF), which is typically around 25 Hz, leading to the perception of a blended color without discernible flicker [Truss 1957].

Imperceptible color vibration can be used to embed information such as a 2D barcode in LCD images that is not perceived by the human eye but detectable by cameras or photosensors [Abe et al. 2020]. An increase in the amplitude of the color vibration facilitates the detection of the vibration by the camera, yet simultaneously renders the flicker more perceptible to humans. The conventional approach to selecting color-vibration pairs is based solely on distances in the color space [Abe et al. 2020; Hattori and Hiraki 2022].

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However, the human eye exhibits inequality in color perception; for instance, even if two colors on the chromaticity diagram are separated by the same distance, two colors closer to blue can be distinguished, but two colors closer to green cannot. The method of efficiently searching for and generating imperceptible color vibrations taking into account the perceptual characteristics of the human eye is not explored.

This paper proposes a method for selecting color vibration pairs with respect to the inequality of color perception based on the MacAdam ellipse [MacAdam 1942] (Fig. 1). The MacAdam ellipse represents the experimentally determined range of color differences that cannot be distinguished by the human eye with respect to a particular color. We conducted an experiment to create color pairs based on this ellipse and confirm the threshold of the color vibration amplitude at which flicker is undetectable by the human eye.

2 METHODS

The MacAdam ellipse $\mathcal{E}_n = \{c_n, \theta_n, a_n, b_n\}$ ($n = 1 \cdots 25$) is defined at 25 points on the xy chromaticity diagram by its center $c_n = [c_{nx}, c_{ny}]$, rotation angle θ_n , and the lengths a_n and b_n of long and short diameter. We choose color pairs $\{p_n^+(r), p_n^-(r)\}$ multiplied by ratio r along the long diameter a_n of these ellipses as color vibration pairs, denoted as $p_n^\pm(r) = [c_{nx} \pm r \cdot a_n \sin \theta_n, c_{ny} \pm r \cdot a_n \cos \theta_n]$. This allows for the selection of color vibration pairs while taking into account the inequality of human color perception.

xy chromaticity diagram is calculated by normalizing the luminance Y of the CIEXYZ color space. Therefore, when displaying colors based on the selected $p_n^\pm$ in practice, it is necessary to complement the luminance. As Y approaches 0, $p_n^\pm$ approaches black, and colors become nearly invisible. In contrast, as Y approaches 1, the brightness of $p_n^\pm$ exceeds the sRGB range. Therefore, we set $Y = 0.4$ and convert xy value to XYZ with $xyY_to_XYZ(xyY)$ from the colour-science library in Python¹. The selected color pairs in CIEXYZ are then converted to the sRGB color system to display the pairs. To convert the colors, we use CIE 1931 2° as an observer function under the D65 illuminant.

3 EXPERIMENT AND RESULTS



Figure 2: Images of 8 colors defined by the MacAdam ellipses used to generate color vibrations in the experiment.

We conducted a user experiment to confirm the threshold of r at which humans perceive flicker. In this experiment, 8 points within the sRGB color space were selected from 25 points within the xy color space defined by the MacAdam ellipse to generate color vibrations. Figure 2 shows the colors of the 8 selected points. We generate color pairs with r values from 1 to around 40, split into 8 varying intervals across colors, and randomly presented these pairs on a screen in front of the participant. Since the color space in sRGB is smaller than the xy color space, some of $p_n^\pm$ cannot be reproduced in the sRGB color space. By omitting these pairs,

we used 46 pairs in this experiment. To detect random responses, 46 single colors without color vibration were also displayed. As a result, each participant saw a total of 92 color patterns.

Ten participants (8 males, 2 females) are asked to respond if they can perceive flicker from the displayed colors. The participants were seated in front of a sRGB color-calibrated LCD display (ColorEdge CG2420-Z, EIZO Inc.) at a distance of 60 cm so that their eye level was in the center of the monitor. Each color pair was displayed as a 15 cm square at 60 cm from the monitor. After each of the five pairs, the participants took a break to look at the black screen. The protocol was approved by Cluster, Inc. Research Ethics Committee, and informed consent was obtained from all participants.

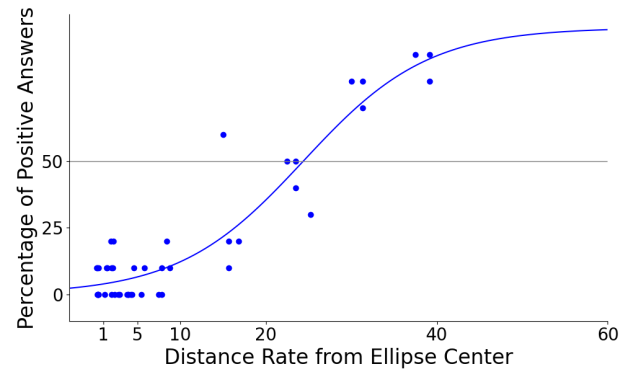


Figure 3: Percentage of responses perceived flicker relative to r . The blue dots indicate the mean, and the blue line shows the perceptual curve fitted by the sigmoid function.

Figure 3 shows the average percentage of responses perceived as flicker compared to r . From the result, the r value of perceiving color vibration 50 % of the time was $r = 24.4$. The results show that perceptual tolerance is approximately $\times 24$ expanded when the two colors alternate in time, in contrast to the MacAdam ellipses, which examined discrimination thresholds when the two colors are juxtaposed spatially.

4 CONCLUSION

This paper presented a method for generating color vibration pairs based on perceptual metrics and a general policy on acceptable amplitudes for pair selection. This method of selecting color pairs can be extended to any color by interpolating MacAdam ellipses at each point. Future studies include the threshold of r relative to the hue value and the adaptation of individual perceptual differences.

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¹Colour 0.4.4 by Colour Developers, <https://zenodo.org/records/10396329>